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Effects of mono- and divalent cations on the structure and thermodynamic properties of polyelectrolyte gels

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Measurements are reported on the effect of monovalent and divalent salts on the swelling behavior and supramolecular structure of sodium polyacrylate gels (NaPA) made by osmotic swelling pressure and small angle neutron scattering measurements. The swelling response of the gels in solutions of ten different monovalent salts is found to be practically identical indicating that the principal effect of monovalent ions is screening the electrostatic repulsion among the charged groups on the polyelectrolyte chains; *i.e.*, chemical differences between the monovalent ions do not play a significant role. Introducing Ca^{2+} ions into the equilibrium NaCl solution results in a sharp volume transition of the gels. The threshold Ca^{2+} ion concentration at which the transition occurs increases with increasing NaCl concentration in the surrounding bath. It is demonstrated that the swelling behavior of NaPA gels exhibits universal properties in solutions containing both NaCl and CaCl_2 . Osmotic swelling pressure measurements reveal that both the second and third virial coefficients decrease with increasing CaCl_2 concentration until the volume transition is reached. The macroscopic measurements are complemented by small angle neutron scattering that reveals the variation of the thermodynamic length scales as the volume transition is approached. The thermodynamic correlation length L increases with increasing CaCl_2 concentration.

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Introduction

Polyelectrolyte gels comprise of charged macromolecules and a solvent which may constitute more than 99% of the total volume.¹ The physical properties (mechanical, thermal, osmotic, electrical, *etc.*) of these materials give rise to diverse phenomenological behaviors, such as capability to swell (up to thousand times relative to the dry volume of the polymer),² exchange mobile ions from an external solution,³ and response to different environmental stimuli (*e.g.*, temperature, solvent quality, pH, electric or magnetic field, ion composition).⁴

It is known that polyelectrolyte gels exhibit reversible volume transition from the swollen to the collapsed state when monovalent ions are replaced by divalent counter-ions. This phenomenon has been demonstrated on both synthetic (*e.g.*, sodium polyacrylate) and biopolymer gels (*e.g.*, DNA).⁵ In weakly cross-linked systems this transition is sharp and can be initiated by a relatively small change in the ionic composition of the equilibrium solution.

Reversible volume changes play a key role in various biological processes such as cellular excitability,^{6,7} tight packaging

of DNA in the cell nucleus and within viruses,⁸ and the formation of membrane-less organelles.⁹ Furthermore, the presence of both mono- and divalent cations is vital for living cells, and the competition between the ions highly influences the state of the system (for example, the transition from resting to excited state of neurons¹⁰).

Polyelectrolyte gels are also widely used in industrial applications (*e.g.*, ion-exchange,³ water purification,¹¹ controlled drug delivery,¹² sensors,^{13,14} artificial muscles¹⁴). Because of their ability to absorb large volume of water, charged cross-linked polymers are used as superabsorbent materials (*e.g.*, in diaper and related products)² as well as in agricultural applications¹⁵ to improve water retention of the soil.

In general, gel swelling of polyelectrolyte gels is governed by several competing effects. The osmotic pressure of the polymer chains and that of the free counter-ions are the main driving forces of gel swelling. In polyelectrolytes, the repulsive electrostatic interaction among the fixed charges on the polymer backbone also play an important role.^{16,17} These effects are balanced by the elasticity of the network. At equilibrium the latter counteracts the other contributions. Addition of salt progressively screens the electrostatic repulsion, which leads to deswelling of the gel. However, the role of residual long-range electric forces within polyelectrolyte gels raises experimental and theoretical challenges and is not fully understood.¹⁸

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It has been found that in the presence of excess salt the swelling degree of weakly cross-linked polyelectrolyte gels is governed primarily by the osmotic pressure of the network polymer, while the osmotic contribution of the free ions is negligible.¹⁹ Structural studies made by anomalous X-ray scattering indicate that inside the gel divalent counter-ions occupy the immediate neighborhood of the charged polymers and reduce the size of the ion cloud around the polyelectrolyte chain.²⁰

The objective of the present study is to describe the swelling behavior of polyelectrolyte gels equilibrated with solutions of ten monovalent salts and in mixtures of mono- and divalent salts. The competition between Na^+ and Ca^{2+} ions is discussed. A further objective is to quantify the effect of Ca^{2+} ions on the thermodynamic interaction and supramolecular structure of the gels in the course of the ion exchange process. To this end osmotic deswelling measurements and small angle neutron scattering (SANS) measurements are made on neutralized sodium polyacrylate (NaPA) gels as a function of the CaCl_2 concentration as the volume transition is approached.

The paper is organized as follows. After a brief description of the materials and methods, results of equilibrium swelling measurements made in solutions of ten monovalent salts as a function of the salt concentration in the equilibrium solution are presented. The competition between Na^+ and Ca^{2+} is investigated by gradually increasing the Ca^{2+} concentration at constant NaCl concentrations. The universal behavior of polyelectrolyte gel swelling is discussed in solutions containing both NaCl and CaCl_2 using scaled variables. Then the results of osmotic swelling pressure measurements made at constant ion concentrations are discussed. The effect of Ca^{2+} ions on the organization of the cross-linked polyelectrolyte chains is determined by SANS. A comparison is made between the longitudinal osmotic moduli determined by SANS and derived from macroscopic osmotic and mechanical observations.

Materials and methods

Salt solutions

Ten solutions of 1:1 electrolytes were made from LiCl, NaCl, KCl, RbCl, CsCl, LiBr, NaBr, KBr, NaI and KI. The monovalent salt concentration was varied in the range 0 mM to 300 mM. Solutions containing both monovalent (1:1) and divalent (2:1) electrolytes were prepared from NaCl and CaCl_2 , at ionic concentrations of 10–200 mM and 0–10 mM, respectively. All materials were purchased from Sigma-Aldrich and used without further purification.

Gel preparation

NaPA gels were synthesized by free-radical copolymerization of sodium acrylate and *N-N'*-methylene-bisacrylamide in aqueous solution. First, 30% (w/w) acrylic acid (Sigma-Aldrich) solution was fully neutralized with stoichiometric amount of sodium hydroxide (Sigma-Aldrich). After neutralization, the cross-linker (*N-N'*-methylene-bisacrylamide) was added. The weight ratio of

the cross-linker to the monomer was 1:750, which corresponds to approximately 1600 monomer units per one cross-linker molecule. The polymerization reaction was initiated by ammonium persulfate (0.7 g L^{-1}). A typical recipe included 9 g acrylic acid, 22 ml deionized water, 5 g NaOH, 0.012 g *N-N'*-methylene-bisacrylamide, and 0.021 g ammonium persulfate. After homogenization, the solution was poured into a silicon mold and kept at 95°C for 1.5 hours in an oven. Subsequently, the gels (cuboid shape: $4 \text{ mm} \times 4 \text{ mm} \times 2 \text{ mm}$) were stored at room temperature for 24 hours to ensure that the reaction was completed. Gels were then removed from the mold and placed in deionized water for 12 hours to remove unreacted materials (e.g., sol fraction). Finally, the gels were dried, and the weight of the dry polymer was measured.

Swelling measurements

Gels were placed in containers with salt solutions of different concentrations. The volume of the equilibrium salt solution was approximately two thousand times greater than that of the dry gel. The solution was refreshed after 24 hours. Before measuring the weight of the swollen gel, excess liquid droplets were gently removed from the surface by a tissue paper. The weights of the swollen gels were measured multiple times, until equilibrium was attained. The volume fraction of the polymer was calculated from the known densities of the components ($\rho_{\text{water}}: 0.997 \text{ g cm}^{-3}$, $\rho_{\text{NaPA}}: 1.22 \text{ g cm}^{-3}$) assuming volume additivity.

Measurements were made on at least three independent gel samples and the mean values are reported.

Swelling pressure measurements

Swelling pressure measurements were made by an osmotic stress technique.²¹ Gel samples were equilibrated with poly(vinyl pyrrolidone) (PVP) solutions (molecular weight: 29 kDa) of known osmotic pressure. A semipermeable membrane was used to separate the gels from the solutions to prevent the diffusion of the PVP molecules into the swollen network.²² After attaining equilibrium (approximately 4–5 days), the polymer concentration was measured in both phases.

In dilute polymer solutions the osmotic pressure is governed by the molecular weight of the polymer (M), and the interaction between the polymer and the solvent. In solutions, in which the chains are molecularly distributed, the osmotic pressure Π_{mix} is given by eqn (1)

$$\Pi_{\text{mix}} = (RT/V_1)(\phi/M + A_2\phi^2 + A_3\phi^3 + \dots) \quad (1)$$

where ϕ is the volume fraction of the polymer ($1/\phi$ is the degree of swelling), A_2 and A_3 are virial coefficients, R is the gas constant, T is the absolute temperature and V_1 is the partial molar volume of the solvent. In general, in polymer solutions the second virial coefficient is a measure of the polymer–solvent interaction. In good solvent condition $A_2 > 0$, and in poor solvent condition $A_2 < 0$. In an ideal (theta) solvent, i.e., in the absence of excluded volume interaction, $A_2 = 0$. The third virial coefficient A_3 is related to the flexibility of the polymer.

In polyelectrolyte solutions ions are expected to affect chain flexibility.

In the semidilute regime the polymer chains overlap, and ϕ/M is negligible. In neutral gels, at equilibrium with the pure solvent, the osmotic mixing pressure of the network chains Π_{mix} is balanced by the elastic pressure Π_{el} ($= -G$, where G is the shear modulus) of the gel, and $\Pi_{\text{gel}} = 0$. Thus, for a freely swollen gel

$$\Pi_{\text{mix}} = -\Pi_{\text{el}} = (RT/V_1)(A_2\phi^2 + A_3\phi^3 + \dots). \quad (2)$$

In polyelectrolyte gels an additional term Π_{ion} exists that arises from the contribution of the ions to the osmotic pressure.^{23,24} However, molecular dynamics simulations indicated that in presence of excess added salt the ionic term is negligible because the electrostatic contribution and the counter-ion excluded volume contribution cancel each other.²⁵ Several experimental studies indicated that expressions derived for neutral polymer solutions satisfactorily describe the osmotic behavior of polyelectrolyte solutions.²⁶

Mechanical measurements

The elastic (shear) moduli G of NaPA gels were determined from uniaxial compression measurements made by a TA.XT2I HR Texture Analyzer (Stable Micro Systems, UK). Measurements were performed on isometric gel cylinders (height = diameter ≈ 1 cm) in equilibrium with salt solutions. G was calculated from the so-called nominal stress, σ (force per unit undeformed cross section), using the relation²⁷

$$G = \frac{\sigma}{\lambda - \lambda^{-2}} \quad (3)$$

where λ is the deformation ratio. Measurements were performed in the deformation range $0.7 < \lambda \leq 1$. Volume change and barrel distortion during the mechanical measurements were not observed.

All measurements were made at 25 ± 0.1 °C.

Small angle neutron scattering (SANS)

SANS measurements were made on the NG3 instrument at incident wavelength 8 Å in the National Institute of Standards and Technology (NIST), Gaithersburg, MD. Gels prepared in D₂O were placed in 2 mm thick sample cells with quartz windows. The measurements were made at three sample-detector distances: 1.3, 4 and 13 m. This configuration spanned the transfer wave vector range $0.002 \text{ Å}^{-1} < q < 0.3 \text{ Å}^{-1}$, where $q = (4\pi/\lambda) \sin(\theta/2)$, and λ and θ are the wavelength of the incident radiation and the scattering angle, respectively. After azimuthal averaging, corrections for incoherent background, detector response and cell window scattering were applied. The ambient temperature during the experiments was 25 ± 0.1 °C. The SANS measurements were made below the threshold Ca²⁺ ion concentration at which volume transition occurs.

Analysis of SANS measurements

For polymer gels composed of flexible chains the SANS intensity can be described as a sum of dynamic and static contributions

$$I(q) = I_{\text{dyn}}(q) + I_{\text{st}}(q) \quad (4)$$

In eqn (4) the first term is governed by scattering from thermodynamic concentration fluctuations, while the second term arises from static structures, which contribute negligibly to the thermodynamics.

For semidilute polymer solutions $I_{\text{dyn}}(q)$ is given by an Ornstein-Zernike function,

$$I_{\text{dyn}}(q) = \Delta\rho^2 \left(\frac{kT\phi^2}{K_{\text{os}}} \right) \frac{1}{1 + q^2\xi^2} \quad (5)$$

where $\Delta\rho$ is the neutron scattering contrast factor between the polymer and solvent, K_{os} is the osmotic compression modulus, ξ is the correlation length defined by the extent of thermal concentration fluctuations and k is the Boltzmann constant.

The scattering response of polyelectrolyte solutions in the presence of added salt can be described by the following empirical expression^{28,29} which contains three adjustable parameters:

$$I_{\text{dyn}}(q) = \Delta\rho^2 \left(\frac{kT\phi^2}{K_{\text{os}}} \right) \frac{1}{(1 + (qL)^2)^{1/2}} \frac{1}{1 + q^2R^2} \quad (6)$$

where L is the correlation length, and R is the cross-sectional radius of the polymer chain. In the overlapping regime, the concentration fluctuations, which give rise to the scattering, correspond to fluctuations in the separation between the polymer molecules.

Both in polymer solutions and gels large-scale structures are present. In such systems the description of the scattering response requires an additional term. In gels, the cross-linking process always leads to formation of static inhomogeneities, which are frozen in by the cross-links. In polyelectrolyte systems large clusters are formed due to molecular chain association (clustering), which cause excess scattering at low q . In general, the scattering from such large objects can be described by a power law

$$I_{\text{st}}(q) = \frac{B}{q^m} \quad (7)$$

where B and m are constants. For objects with smooth interface $m = 4$ (Porod regime), whereas $3 < m < 4$ characterizes rough surfaces.

The total scattering intensity $I(q)$ is the sum of eqn (6) and (7)

$$\begin{aligned} I(q) &= I_{\text{dyn}}(q) + I_{\text{st}}(q) \\ &= \Delta\rho^2 \left(\frac{kT\phi^2}{M_{\text{os}}} \right) \frac{1}{\sqrt{1 + (qL)^2}} \frac{1}{1 + q^2R^2} + \frac{B}{q^m} \end{aligned} \quad (8)$$

where M_{os} is the longitudinal osmotic modulus.³⁰ In polymer gels the scattering response from the thermodynamic fluctuations is governed by M_{os} , which replaces K_{os} in eqn (6).

Results and discussion

Effect of salt concentration on the equilibrium swelling degree of NaPA gels

Fig. 1 shows the swelling degree of nominally identical NaPA gels measured as a function of the salt concentration in solutions of ten different monovalent salts (LiCl, NaCl, KCl, RbCl, CsCl, LiBr, NaBr, KBr, NaI, and KI). The salt concentration was varied in the equilibrium bath and the degree of swelling of the gels was measured until equilibrium was attained. The swelling degree smoothly and gradually decreases with increasing salt concentration from $1/\varphi \approx 500$ (in de-ionized solution) to ≈ 30 (in 300 mM salt solution). All data fall on the same curve indicating that in monovalent salt solutions the gels exhibit similar behavior; *i.e.*, chemical differences between monovalent ions do not play a significant role. This result implies that the main effect of monovalent ions is screening the electrostatic repulsion among the charged groups on the polyelectrolyte chains. This finding is consistent with previous results reported for limited number of monovalent salts.^{26,31} In contrast, certain polyelectrolyte systems, such as protein solutions and gels, show significant 'ion specific' effects.³² The discrepancy was attributed to the nature of the polymer backbone. It was suggested that direct interactions of ions with non-polar surface are the source of the so-called Hofmeister effect.³³

In the inset in Fig. 1 are plotted the same data in a double logarithmic representation. The swelling degree exhibits a power law dependence on the salt concentration $\frac{1}{\varphi} \propto [c]^n$, where $n = -0.47 \pm 0.03$. This behavior resembles the variation of the electrostatic screening length λ_D with the concentration of added monovalent salt ions, $\lambda_D \sim (\ell_B c_{\text{salt}})^{-1/2}$, where ℓ_B is the Bjerrum length.¹⁸

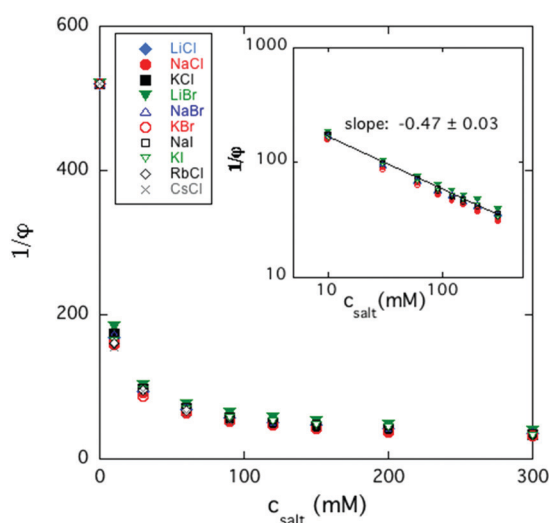


Fig. 1 Variation of the equilibrium swelling degree of NaPA gels with the concentration of various monovalent salts. Inset shows the same data in double logarithmic representation. The reproducibility of the swelling measurements was $<5\%$. For clarity error bars are not shown in the figure.

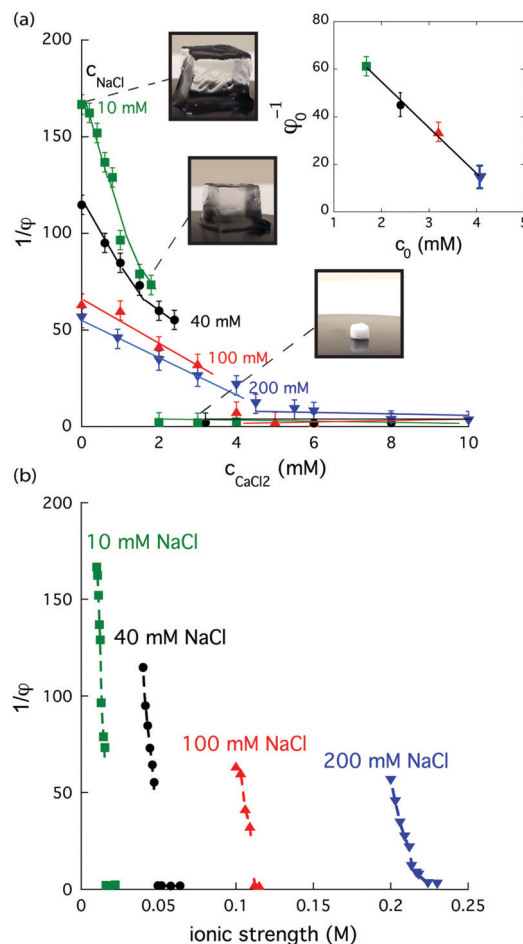


Fig. 2 (a) The equilibrium swelling degree of NaPA gels as a function of the concentration of CaCl_2 in the surrounding solution at different NaCl concentrations. Continuous lines through the swelling data are guides to the eye. (b) Equilibrium swelling degree of NaPA gels as a function of the ionic strength of the equilibrium solution.

Fig. 2a illustrates the effect of Ca^{2+} ions on the swelling degree of gels swollen in NaCl solutions at different NaCl concentrations (10 mM, 40 mM, 100 mM and 200 mM NaCl). In the absence of Ca^{2+} ions the gels swell less with increasing the NaCl concentration. It can be seen that the swelling degree progressively decreases with increasing CaCl_2 concentration in the surrounding NaCl solution, and at a threshold CaCl_2 concentration c_0 a sudden volume transition occurs. The inset in Fig. 2a shows the relationship between the polymer volume fraction φ_0 at which the transition occurs and c_0 . (φ_0 is the volume fraction of the polymer in the swollen gel just below the volume transition.) It can be seen that the swelling degree $1/\varphi_0$ linearly decreases with c_0 indicating that to induce volume transition more CaCl_2 is needed when the NaCl concentration is greater in the external solution reflecting the competition between the mono- and divalent cations for the negatively charged carboxylic groups on the polymer chains. At high NaCl concentration ($c_{\text{NaCl}} > 200$ mM, not shown here) $1/\varphi$ continuously decreases with increasing CaCl_2 content and noticeable volume transition cannot be observed. Fig. 2b shows the same

swelling data as a function of the ionic strength of the equilibrium solution. It is clear from both Fig. 2a and b that the volume transition occurs at a lower degree of swelling with increasing (i) NaCl concentration, and (ii) ionic strength. The reversible nature of the volume transition observed in the present gels indicates that binding of Ca^{2+} ions is not permanent. Therefore, it is reasonable to assume that Ca^{2+} ions increase the attractive interaction among the polymer molecules favoring cluster formation that ultimately leads to volume transition.

To take into account the effect of monovalent ion concentration on the swelling curves we introduced the reduced polymer volume fraction ϕ/ϕ_0 and the reduced salt concentration c/c_0 to replace ϕ and c , respectively. In Fig. 3 we replotted the swelling data in terms of these variables. In this representation all data points collapse on a single master curve indicating that the swelling behavior of NaPA gels in solutions containing NaCl and CaCl_2 exhibits universal behavior.

We note that the universal behavior described above is expected to be observed in homologous gels, *i.e.*, gels having similar chemical composition, topology, *etc.*

Effect of calcium ions on the thermodynamic properties of NaPA gels

In the previous section we have discussed the behavior of freely swollen NaPA gels in equilibrium with NaCl solutions containing increasing amounts of Ca^{2+} ions. To gain information on the effect of the Ca^{2+} ions on the elastic modulus and thermodynamic interactions in the gels we made osmotic deswelling measurements at constant ion concentration and composition.

Fig. 4 shows the variation of the elastic shear modulus G as a function of the swelling degree measured in NaCl solutions containing different amounts of CaCl_2 . In the double logarithmic representation all data points fall on a single master curve.

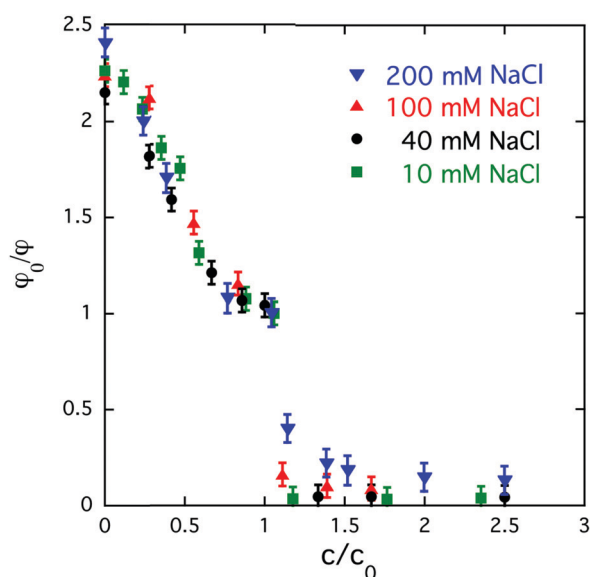


Fig. 3 ϕ/ϕ_0 and c/c_0 plots for NaPA gels in NaCl solutions of different concentrations.

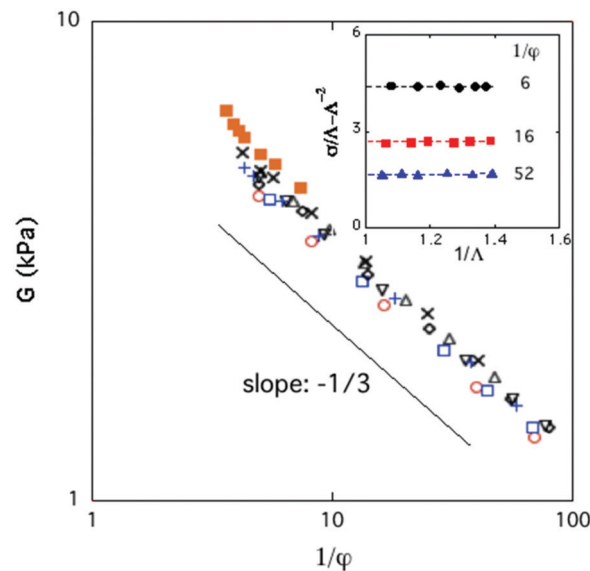


Fig. 4 Variation of the elastic modulus as a function of the swelling degree $1/\phi$ of NaPA gels in equilibrium with salt solutions. Symbols: $\square$ 10 mM NaCl solution, $\circ$ 40 mM NaCl solution, ∇ 100 mM NaCl solution, $+$ 40 mM NaCl + 0.2 mM CaCl_2 , $\triangle$ 40 mM NaCl + 0.5 mM CaCl_2 , $\times$ 40 mM NaCl + 0.8 mM CaCl_2 , $\blacksquare$ 40 mM NaCl + 1 mM CaCl_2 . Inset shows typical $\frac{\sigma}{\Delta - \Delta^{-2}}$ vs. $1/\Delta$ plots for the 40 mM NaCl + 0.5 mM CaCl_2 gel sample at three swelling degrees.

This results implies that the elastic modulus is a function of the swelling degree only, *i.e.*, the ionic environment, including the Ca^{2+} ions, does not modify the cross-link density of the gel. The observed concentration dependence of G is consistent with the prediction of the theory of rubber elasticity, $G = G_0\phi^{1/3}$, where the constant G_0 is proportional to the number of elastic chains in the gel.²⁷

Fig. 5 shows the variation of the normalized swelling degree as a function of the normalized concentration c/c_0 . In the inset is plotted $\Pi_{\text{mix}} (= \Pi_{\text{gel}} + G)$ determined from swelling pressure and uniaxial compression measurements made on gels in 40 mM NaCl solution with different CaCl_2 concentrations. At constant ion composition Π_{mix} gradually decreases with increasing CaCl_2 content indicating that Ca^{2+} ions screen the electrostatic repulsion between the charged groups on the polymer backbone. Above a threshold Ca^{2+} concentration the osmotic pressure vanishes and volume transition occurs (red dashed curve in the inset). In the present NaPA gel volume transition occurs at ≈ 2.5 mM CaCl_2 concentration.

In the figure are also displayed the fits of eqn (9) to the $\Pi_{\text{mix}} (= \Pi_{\text{gel}} + G)$ data

$$\Pi_{\text{mix}} = \frac{RT}{V_1} (A_2\phi^2 + A_3\phi^3) \quad (9)$$

It can be seen that eqn (9) satisfactorily describes the experimental data over the whole concentration range explored in the present study.

In Fig. 6 are shown the virial coefficients obtained from the fits of eqn (9) as a function of the normalized CaCl_2

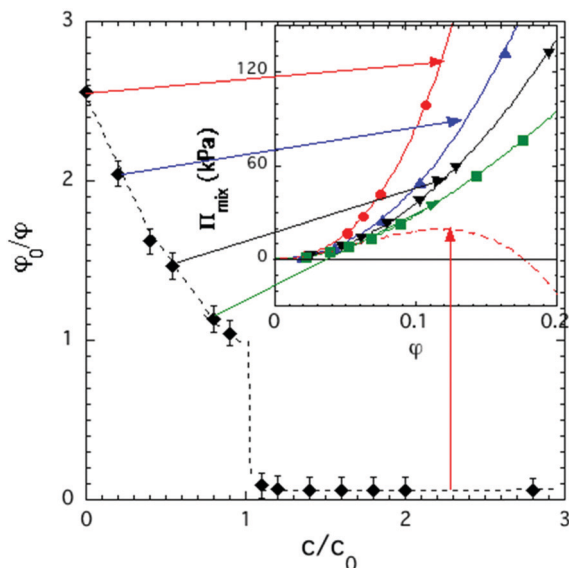


Fig. 5 Normalized swelling degree ϕ_0/ϕ vs. normalized CaCl_2 concentration c/c_0 for NaPA gel in 40 mM NaCl solution. Inset: Π_{mix} vs. ϕ plots. Continuous lines through the data points are fits of eqn (9) to the experimental data. The arrows directed to the curves in the inset show the Π_{mix} vs. ϕ plots measured at selected values of c/c_0 as indicated in the main figure.

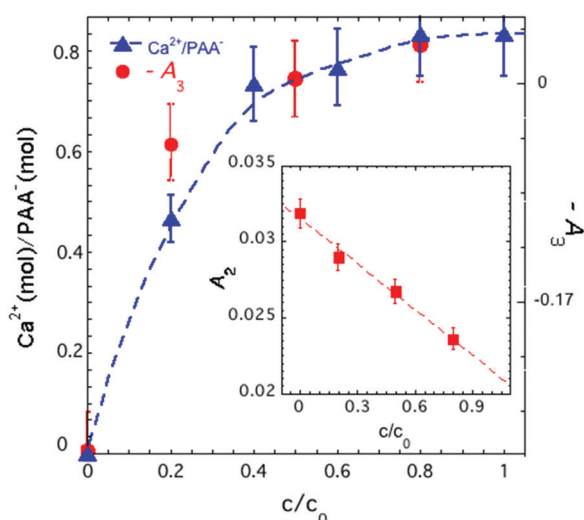


Fig. 6 Number of Ca^{2+} ions per number of monomers in the swollen gel (blue triangles), A_2 (inset) and A_3 (main figure, red dots) as a function of the normalized concentration c/c_0 .

concentration in the equilibrium 40 mM NaCl solution. The second virial coefficient A_2 (inset, red squares) decreases weakly and linearly with increasing c/c_0 , while A_3 (red circles) monotonically but non-linearly decreases with the calcium concentration and becomes negative when $c/c_0 > 0.5$ (main figure, right axis). This finding indicates that the continuous decrease of the virial coefficients leads to a volume transition.

In Fig. 6 is also displayed the number of Ca^{2+} ions divided by the number of monomer units in the gel as a function of c/c_0 . The Ca^{2+} concentration was determined by flame spectroscopy

(Galbraith Laboratories, Nashville, TN²⁶). It can be seen that A_3 closely follows the Ca^{2+} content of the gel, and implies that A_3 linearly decreases as the amount of Ca^{2+} ions increases inside the gels.

It is worth pointing out that in the present system λ_D gradually approaches ℓ_B as the salt concentration increases in the external solution. Specifically, $\ell_B \approx 0.71$ nm and λ_D decreases from 3 nm (in 10 mM NaCl) to 0.7 nm (in 180 mM NaCl). At higher NaCl concentrations $\lambda_D < \ell_B$. Since the seminal Debye-Hückel theory is valid only when $\lambda_D \gg \ell_B$, this simple model is insufficient to capture the thermodynamic behavior of this system. Furthermore, according to Manning, ion condensation occurs if $\frac{\ell_B}{d} > 1$, where d is the distance between the charge groups on the polymer chain.³⁴ This condition is satisfied for the present system, since $d \approx 0.3$ nm. Thus, the residual long-range electrostatic interactions may give rise to complex thermodynamic behavior at multiple length scales (including temporary dipoles, quadrupoles and larger structures).¹⁸

The role of Ca^{2+} ions beyond the transition is even less clear. If Ca^{2+} ions are adsorbed on the polyanion, several changes may occur due to the electrostatic screening between the charged groups on the polymer backbone. Replacement of monovalent counter-ions by divalent ions may induce charge redistribution in the ion cloud and at high polymer volume fraction ion adsorption may favor bridge formation between the polymer molecules. Since divalent ions satisfy two negative charges on the polymer chain, the replacement of loosely bound Na^+ ions by Ca^{2+} ions are expected to reduce the repulsion between the polyelectrolyte chains. Divalent/monovalent ion exchange should also decrease the rigidity of the chains which is consistent with both experimental observations and molecular dynamics simulations.^{35,36}

Structural changes induced by $\text{Ca}^{2+}/\text{Na}^+$ ion exchange

To obtain information on the structural changes occurring in NaPA gels with increasing Ca^{2+} concentration we made SANS measurements.

Fig. 7 shows that the scattering profiles display similar properties. The overall scattering intensity gradually increases with increasing Ca^{2+} concentration. In the low q region, $I(q)$ strongly decreases with increasing q indicating the presence of large clusters the size of which exceeds the resolution of the SANS experiment. The slope $m \sim -3.6$ is characteristic of rough surfaces. Cluster formation has been reported for many polyelectrolyte solutions and gels.^{5,29,35,37} In gels the polymer rich regions are dispersed in a continuous matrix, and connected through regions of lower polymer volume fraction, in which the chains are stretched due to residual electrostatic interactions among the charged groups on the backbone. The present results show that the low q feature of the scattering profile is practically unaffected by the Ca^{2+} concentration. In the intermediate q -range the q dependence of the scattering signal becomes weaker $I(q) \sim q^{-1}$. In the highest q region, the scattering response is governed by the local geometry of the

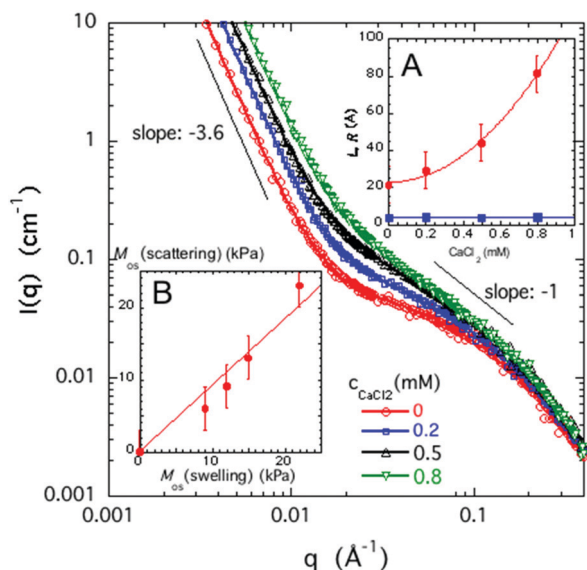


Fig. 7 SANS profiles of NaPA gels ($\phi = 0.04$) swollen in 40 mM NaCl containing increasing amounts of CaCl_2 as indicated in the figure. Continuous curves: fit of eqn (8). Inset (A): Variation of the correlation length L (red dots) and the cross-sectional radius R (blue squares) as a function of the CaCl_2 concentration. Inset (B): $M_{\text{os}}(\text{scattering})$ vs. $M_{\text{os}}(\text{swelling})$ for NaPA gels.

polymer chain. The position of the shoulder is defined by the cross-sectional radius of the polymer molecule. The continuous curves through the data points are the least squares fits of eqn (8) to the SANS data. Inset (A) in Fig. 7 indicates that the correlation length L increases with increasing Ca^{2+} concentration from approximately 19 Å (CaCl_2 free solution) to 81 Å (0.8 mM CaCl_2 in 40 mM NaCl). The increase of L implies chain aggregation. Ca^{2+} ions screen the electrostatic repulsion between the polymer molecules and enhance interchain association. The q^{-1} dependence of the SANS intensity suggests that the aggregation process favors chain alignment. Similar behavior has been reported for semidilute solutions of different polymers.²⁹ In the present experiment the volume of the gel was constant ($\phi = 0.04$) and the rapid increase of L reflects the onset of phase separation. The values obtained for the cross-sectional radius of the polymer chain R are close to 4 Å and practically independent of the CaCl_2 concentration.

Comparison of the osmotic modulus determined from osmotic swelling pressure measurements and SANS

Counter-ions influence both the thermodynamic properties of the gel and the organization of the network chains. SANS indicates that Ca^{2+} ions primarily affect the scattering intensity in the intermediate q range. Moreover, inset (A) in Fig. 7 shows that L increases with increasing CaCl_2 concentration.

The thermodynamic contribution to the SANS signal can be estimated from the first term of eqn (8). According to general thermodynamic considerations, the intensity scattered by osmotic concentration fluctuations $I_{\text{dyn}}(0)$ is governed by M_{os} . In Fig. 7 the increase in the scattering intensity with increasing Ca^{2+} ion concentration reflects the decrease of the osmotic

modulus M_{os} ; *i.e.*, as the system moves towards the transition the gel becomes softer.

Measurement of the osmotic swelling pressure Π_{gel} and the shear modulus provides an independent estimate of M_{os} .³⁸

$$M_{\text{os}}(\text{swelling}) = \phi \frac{\partial \Pi_{\text{gel}}}{\partial \phi} + \frac{4}{3}G \quad (10)$$

Inset (B) in Fig. 7 shows that $M_{\text{os}}(\text{swelling})$ from macroscopic measurements is in reasonable agreement with $M_{\text{os}}(\text{scattering})$ determined from SANS at $I(q) = 0$. Similar agreement between osmotic and scattering results has also been reported for neutral gels.^{39–41} These results indicate that the SANS response of the present NaPA gels can be decomposed into a dynamic and a static component. The former is identified with the thermodynamic concentration fluctuations and is simply related to the macroscopic osmotic swelling pressure and elastic modulus of the swollen network. The static component of the total scattered intensity is the result of local variations of the cross-linking density, which introduce permanent elastic constraints.

Conclusions

NaPA gels exhibit universal swelling behavior in the presence of monovalent salts. The effect of ten different monovalent salts (LiCl, NaCl, KCl, RbCl, CsCl, LiBr, NaBr, KBr, NaI, and KI) was investigated over the concentration range 0–300 mM. In all systems the swelling degree of the gels decreases monotonically from above 500 (in a salt free solution) to about 30 (in 300 mM monovalent salt solutions). The absence of appreciable differences between different monovalent salts indicates that gel deswelling is primarily caused by screening of the electrostatic interaction among the charged groups on the polymer chains.

In salt solutions containing both Na^+ and counter-ions a sharp volume transition is observed as the concentration of the Ca^{2+} ions increases. In the present system the volume transition induced by divalent/monovalent ion exchange is fully reversible and exhibits universal properties in terms of scaled variables ϕ/ϕ_0 and c/c_0 .

Osmotic swelling pressure measurements made on gels with increasing Ca^{2+} ion concentrations indicate that Π_{mix} gradually decreases and above the volume transition it is vanished. Correspondingly, both the second and third virial coefficients decrease with increasing CaCl_2 content.

The SANS profiles of NaPA gels containing different amounts of CaCl_2 display similar general features. In the low q region $I(q)$ is dominated by scattering from large clusters, the size of which exceeds the resolution of the SANS experiment. In the intermediate q range the scattering intensity increases with increasing CaCl_2 concentration. Close to the transition (in the swollen state) the intensity varies approximately as q^{-1} , indicating that the scattering structures are linear. The correlation length L of the concentration fluctuations increases with increasing CaCl_2 concentration as the volume transition is approached.

Conflicts of interest

There are no conflicts to declare.

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